

Interactions of Hippocampal Category Learning with Prior Knowledge

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Abstract

There is increasing evidence that the hippocampus plays a central role in rapid novel category learning, where prior knowledge is unrelated to the new category structure. In natural settings, however, category learning often builds on rich, structured semantic spaces shaped by prior experience. Here we investigated how humans integrate new category information with existing knowledge during learning and provide a hypothesis for how the hippocampus builds from prior knowledge to support this integration. We created categories of known stimuli (e.g., daily foods) that were either congruent or incongruent with the similarity structure of their word embeddings. In simulations using a neural network model of the hippocampus, we demonstrated how different pathways within the hippocampus differentially benefit from congruence. Preliminary results from human behavior were consistent with the model's account. Together, the work advances our understanding of the neural computations that may support the interaction between prior knowledge and new learning.

Keywords: category learning; semantic knowledge; hippocampus; memory

Introduction

The ability to categorize objects and events is fundamental to cognition, enabling generalization across novel experiences. A growing body of work suggests that the hippocampus plays a central role in novel category learning (e.g., Kumaran & McClelland, 2012; Schapiro et al., 2017; Mack et al., 2017). These accounts propose that the hippocampus supports learning processes that balance pattern separation and integration, enabling both the encoding of individual exemplars and the extraction of shared category structure. However, this work has primarily relied on artificial stimuli, in which category learning is largely independent of prior knowledge.

In contrast, real-world category learning operates over rich semantic spaces shaped by existing knowledge and beliefs. For example, when learning

which foods are high in a novel nutrient, individuals can leverage prior knowledge about food. Similarly, when learning to categorize people by social roles, individuals may draw on existing knowledge of the warmth and competence dimensions of social perception (Fiske et al., 2007). Indeed, prior work shows that prior knowledge facilitates category learning, leading to faster acquisition and improved generalization for knowledge-consistent categories (Murphy & Allopenna, 1994; Kaplan & Murphy, 2000; Kumaran et al., 2016).

A central open question, however, is how prior knowledge interacts with hippocampal learning systems to shape category learning. We address this question using the C-HORSE framework (Schapiro et al., 2017; Sučević & Schapiro, 2023), a biologically grounded model of hippocampal contributions to structure learning. C-HORSE assigns distinct computational roles to two hippocampal pathways: the trisynaptic pathway (TSP), which builds sparse, pattern-separated representations suited for exemplar memory, and the monosynaptic pathway (MSP), which builds overlapping representations that extract category structure (Sučević & Schapiro, 2023). Here, we extend C-HORSE by incorporating rich semantic prior knowledge via vector representations of natural objects (e.g., food, traits) as model inputs (Singh, Richie, & Bhatia, 2022) to explore how the pathways of the hippocampus may build new categories from prior knowledge.

Methods & Results

Our modeling framework comprised two single-hidden-layer neural network experts, each incorporating properties of hippocampal subfield circuitry (**Figure 1**). The TSP expert featured a larger hidden layer, sparse connectivity, and k-Winner-Take-All inhibition, whereas the MSP expert had a smaller hidden layer, full connectivity, and no inhibitory mechanism. Both experts generate predictions for category learning and exemplar recognition.



We simulated two learning environments by manipulating prior knowledge congruency (**Figure 2**): category labels were assigned to align with k-means clustering of item embeddings ($k = 2$; congruent) or randomly (incongruent). Both the MSP and TSP experts were trained for 500 epochs across multiple runs. We found that the MSP took advantage of prior knowledge to better facilitate rapid learning, whereas the pattern-separation capabilities of the TSP better supported learning when prior knowledge was incongruent (**Figure 2**).

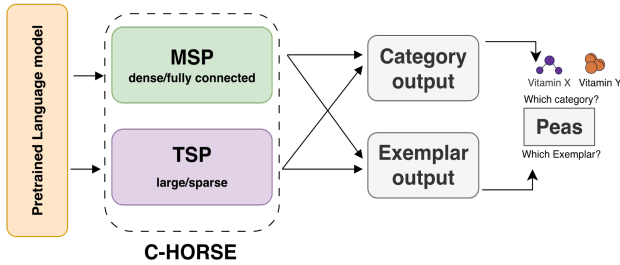


Figure 1. Representations from a pretrained language model are processed by two parallel hippocampal pathway experts: a dense, fully connected MSP and a larger, sparse TSP. Outputs from both pathways are combined to support category and exemplar judgments in a category learning task (e.g., learning whether “peas” are high in a newly introduced vitamin X or Y).

To examine how these patterns manifest in human behavior, we conducted a pilot study in both food and social domains. We aimed to manipulate the relative engagement of MSP and TSP using interleaved versus blocked presentation of category exemplars, as prior work has shown that the MSP requires interleaved exposure whereas the TSP can handle blocked exposure (Zhou et al., 2023). In the food domain, participants ($N = 100$) learned to classify 60 food items as high in nutrient X or Y in a novel categorization task using real-world stimuli. Participants were randomly assigned to one of four conditions in a 2×2 design: Category labels were constructed to either align with (congruent condition) or conflict with (incongruent condition) the underlying structure of the stimuli, and training was conducted in either an interleaved or blocked format. Each food item was presented once during the training phase. Following training, participants completed a test phase that included 20 old items and 20 semantic foils.

In the social domain, we conducted a parallel 2×2 pilot study ($N = 80$) using warmth and competence trait adjectives as stimuli. Participants learned which traits were suitable for Job A versus Job B. In the congruent condition, high-warmth traits were assigned to one job and high-competence traits to the

other, consistent with prior social knowledge. In the incongruent condition, traits were randomly shuffled across jobs. Following training, participants completed a test phase that included 20 old traits and 40 new traits.

We observed a main effect of prior knowledge across both domains: participants were more accurate and faster when prior knowledge was congruent with new learning (Food: accuracy = 0.91 vs. 0.59; RT = 1067 vs. 1351 ms. Social: accuracy = 0.86 vs. 0.51; RT = 1026 vs. 1129 ms; $ps < .001$). The interaction with curriculum was numerically in the predicted direction, but not significant in these pilot samples: Within the congruent condition, interleaved presentation was associated with faster responses for correct old vs. new items in the food domain (food: RT = 917 vs. 1158 ms, $p = .21$), whereas within the incongruent condition, blocked presentation was associated with faster responses in both domains (food: RT = 1200 vs. 1406 ms, $p = .25$; social: RT = 1252 vs. 1548 ms, $p = .55$).

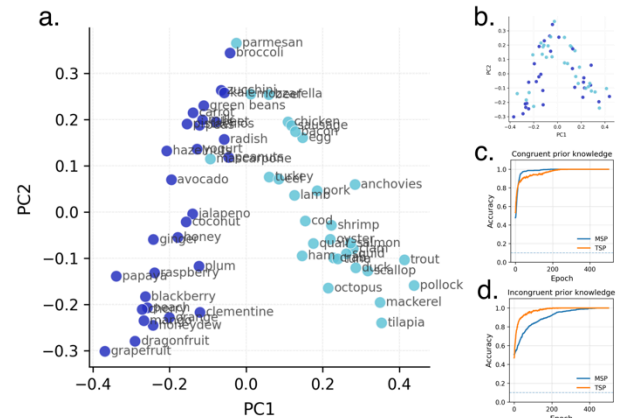


Figure 2. (a–b) Representational geometry under congruent vs. incongruent conditions. (c–d) Learning dynamics of MSP (blue) and TSP (orange).

Conclusion

We introduce a framework for characterizing how prior knowledge interacts with new category learning in the hippocampus. Model simulations predicted that the MSP should benefit from prior congruent knowledge whereas the TSP better supports learning when prior knowledge is incongruent. In a pilot sample testing these predictions behaviorally, we found that congruent prior knowledge broadly facilitates new learning, as in prior work. The curriculum manipulation, aiming to partially dissociate the contributions of the MSP and TSP pathways, generated numerical patterns consistent with the idea that different pathways differentially benefit from prior knowledge congruence, but more data are needed for a full assessment.

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